Estimation

1 Setup and Notation

1.1 Preferences and Production

Worker i's indirect utility for firm j is

$$U_{ij} = \delta_j - \tau d_{ij} + \varepsilon_{ij}, \quad \delta_j \equiv \eta \ln w_j + \xi_j,$$

where ε_{ij} is i.i.d. Gumbel, d_{ij} is commuting distance, $\eta > 0$ is wage sensitivity, and $\tau > 0$ is commuting disutility. Production is $R_j = A_j H_j^{1-\alpha}$, where $H_j = L_j Q_j$ is efficiency units of labor, Q_j is average skill, and $\alpha \in (0, 1)$.

1.2 Skill Specification

We consider two skill specifications. All other notation ($\bar{q}_j$, δ_j, τ, α, η, γ, v_i, e_i, G_e, Q_j) is common to both.

Specification A (additive). Worker skill is linear in observables:

$$q_i = \gamma_0 + \gamma_1 v_i + e_i, \quad e_i \sim N(0, \sigma_e^2),$$

where v_i is an observed skill proxy, e_i is unobserved skill, and γ_0 is a location parameter. Average skill at firm j is $Q_j = \frac{1}{L_j} \sum_{i:y_i=j} (\gamma_0 + \gamma_1 v_i + e_i)$. Acceptance requires $q_i \geq \bar{q}_j$, i.e., $e_i \geq \bar{q}_j - \gamma_0 - \gamma_1 v_i$.

Standardized reparametrization. The match probabilities depend on skill only through $\Phi\left(\frac{\bar{q}_j - \gamma_0 - \gamma_1 v_i}{\sigma_e}\right)$. Define the standardized parameters

$$\tilde{q}_j \equiv \frac{\bar{q}_j - \gamma_0}{\sigma_e}, \qquad \tilde{\gamma} \equiv \frac{\gamma_1}{\sigma_e},$$

so that $t_j(v_i) = \tilde{q}_j - \tilde{\gamma}\, v_i$ in standardized units. The micro data identify $(\tilde{\gamma}, \{\tilde{q}_j\})$ and the preference parameters $(\{\delta_j\}, \tau)$. The structural parameters (γ_0, σ_e)—and hence $\gamma_1 = \tilde{\gamma}\, \sigma_e$—require macro/production data (Section **??**).

Specification M (multiplicative, project description). Log skill is linear:

$$\ln q_i = \gamma_0 + \gamma_1 v_i + e_i, \quad e_i \sim N(0, \sigma_e^2),$$

so $q_i = \exp(\gamma_0 + \gamma_1 v_i + e_i)$. Average skill is $Q_j = \frac{1}{L_j} \sum_{i:y_i=j} \exp(\gamma_0 + \gamma_1 v_i + e_i)$. Acceptance requires $q_i \geq \bar{q}_j$, i.e., $e_i \geq \ln \bar{q}_j - \gamma_0 - \gamma_1 v_i$.

Standardized reparametrization. Match probabilities depend on skill only through $\Phi\left(\frac{\ln \bar{q}_j - \gamma_0 - \gamma_1 v_i}{\sigma_e}\right)$. Define

$$\tilde{q}_j \equiv \frac{\ln \bar{q}_j - \gamma_0}{\sigma_e}, \qquad \tilde{\gamma} \equiv \frac{\gamma_1}{\sigma_e},$$

so that $t_j(v_i) = \tilde{q}_j - \tilde{\gamma}\, v_i$. As in Specification A, the micro data identify $(\tilde{\gamma}, \{\tilde{q}_j\})$ and $(\{\delta_j\}, \tau)$. The structural parameters (γ_0, σ_e) require macro data, but the identification argument differs from Specification A because γ_0 cancels from the A_j-free moment (Section **??**).

Under both specifications, the screening FOC yields

$$w_j = \mathrm{MRPL}_j(\bar{q}_j) = \bar{q}_j \cdot A_j(1-\alpha)(Q_j L_j)^{-\alpha},$$

which can be rewritten as $\bar{q}_j/Q_j = w_j/((1-\alpha)R_j/L_j)$. In logs:

$$\ln \bar{q}_j = \ln w_j - \ln A_j - \ln(1-\alpha) + \alpha \ln(Q_j L_j). \tag{1}$$

1.3 Match Probabilities

Define the threshold function $t_j(v_i) \equiv \tilde{q}_j - \tilde{\gamma}\, v_i$ under both specifications (where $\tilde{q}_j = (\bar{q}_j - \gamma_0)/\sigma_e$ under A and $\tilde{q}_j = (\ln \bar{q}_j - \gamma_0)/\sigma_e$ under M). Then the match probability has a unified form:

$$\sigma_j(v_i, d_i; \theta) = \sum_{O \subseteq \mathcal{J},\, j \in O} \left[\Phi\left(\min_{k \notin O} t_k(v_i) \right) - \Phi\left(\max_{k \in O} t_k(v_i) \right) \right] \frac{\exp(\delta_j - \tau d_{ij})}{1 + \sum_{k \in O} \exp(\delta_k - \tau d_{ik})}. \tag{2}$$

Ordering simplification. Order firms so that $t_{o^{-1}(1)}(v_i) \le t_{o^{-1}(2)}(v_i) \le \cdots \le t_{o^{-1}(J)}(v_i)$ for a representative v_i (i.e., by $\tilde{q}_j$). With conventions $t_{o^{-1}(0)} \equiv -\infty$ and $t_{o^{-1}(J+1)} \equiv +\infty$:

$$\sigma_j(v_i, d_i; \theta) = \sum_{k=0}^{J} \mathbf{1}[t_j \le t_{o^{-1}(k)}] \, \left(\Phi(t_{o^{-1}(k+1)}(v_i)) - \Phi(t_{o^{-1}(k)}(v_i)) \right) \frac{\exp(\delta_j - \tau d_{ij})}{1 + \sum_{m=1}^{k} \exp(\delta_{o^{-1}(m)} - \tau d_{i,o^{-1}(m)})}. \tag{3}$$

Parameter partition. The match probabilities (**??**)–(**??**) depend only on the *standardized* parameters:

$$\theta_{\text{micro}} = \big(\underbrace{\tilde{\gamma}, \tau}_{\text{global}}, \quad \underbrace{(\delta_j, \tilde{q}_j)_{j=1}^{J}}_{\text{firm fixed effects}} \big).$$

The *structural* parameters $(\gamma_0, \sigma_e, \alpha, \eta)$ enter only through the macro/production moments. The full parameter vector is

$$\theta = \big(\underbrace{\tilde{\gamma}, \tau, \gamma_0, \sigma_e, \alpha, \eta}_{\text{global}}, \quad \underbrace{(\delta_j, \tilde{q}_j)_{j=1}^{J}}_{\text{firm fixed effects}} \big).$$

Given (γ_0, σ_e), the structural skill parameters are recovered as $\gamma_1 = \tilde{\gamma}\,\sigma_e$ and $\bar{q}_j = \gamma_0 + \sigma_e \tilde{q}_j$ (Spec. A) or $\ln \bar{q}_j = \gamma_0 + \sigma_e \tilde{q}_j$ (Spec. M). In the multi-market extension, $\theta = (\theta_G, (\theta_m)_{m=1}^{M})$ where $\theta_G = (\tilde{\gamma}, \tau, \gamma_0, \sigma_e, \alpha, \eta)$ and $\theta_m = (\delta_{j_m}, \tilde{q}_{j_m})_{j_m=1}^{J_m}$.

2 The Hybrid MLE/GMM Objective

Following Grieco, Murry, Pinkse & Sagl (2025, "CLEER"), the estimator combines micro-level match data with macro-level (firm-level) moment restrictions. Our context differs from theirs in that we have two sets of firm-level fixed effects—δ_j and $\bar{q}_j$—instead of only δ_j.

The overall objective function is

$$\boxed{\min_{\theta} \quad \underbrace{-\sum_{i=1}^{N} \ln \sigma_{y_i}(v_i, d_i; \theta)}_{\text{Micro: (negative) log likelihood}} \quad + \quad \underbrace{\frac{1}{2} \hat{m}(\theta)' \hat{W}\, \hat{m}(\theta)}_{\text{Macro: GMM penalty}}} \tag{4}$$

where $y_i \in \{0, 1, \ldots, J\}$ denotes worker i's observed match.

2.1 Micro Component: Likelihood of Matches

The micro component is the (negative) log likelihood of the observed matches:

$$\mathcal{L}(\theta) = -\sum_{i=1}^{N} \ln \sigma_{y_i}(v_i, d_i; \theta).$$

Micro likelihood vs. micro GMM. Two interpretations of the micro component are possible, following Grieco et al. (2025, Section 6.1):

(a) **MLE form.** Use $-\sum_i \ln \sigma_{y_i}$ directly as in (**??**). This exploits the global concavity of the logit kernel in (δ, τ) conditional on $(\bar{q}, \gamma)$ and yields efficient estimates when the micro sample is the dominant source of identification.

(b) **GMM form (score-based).** Replace the likelihood with a quadratic form in its score:

$$\frac{1}{2}\left(\sum_i \sum_{j=1}^{J} (y_{ij} - \sigma_j(v_i, d_i; \theta))\, \nabla_\theta \ln \frac{\sigma_j(v_i, d_i; \theta^{(1)})}{\sigma_0(v_i, d_i; \theta^{(1)})}\right)' \hat{W}_\mu \left(\sum_i \sum_{j=1}^{J} (\cdots)\right),$$

evaluated at a first-step estimate $\theta^{(1)}$. As discussed in Grieco et al. (2025, Section 6.1), this GMM version is asymptotically equivalent to the MLE form when $\hat{W}_\mu$ and $\hat{W}_L$ are chosen to reproduce the norming of the likelihood score (their equations (20)–(22)). However, it replaces the computationally tractable logit kernel with a more complicated loss surface that is more prone to local optima.

Open Question: Which form to use for the micro component? The MLE form (a) is computationally more tractable (globally concave in δ given $\bar{q}, \gamma$). The GMM form (b) avoids computing $\ln \sigma$ and may be preferable for a unified GMM framework, but Grieco et al. note it increases local optima. Current simulations use MLE.

Mixed data likelihood (Grieco et al. eq. (7)–(8)). When the micro sample covers only a subset $I_m \subseteq N_m$ of workers in market m, the full data likelihood combines micro and macro terms:

$$\log \hat{L}(\theta) = \underbrace{\sum_m \sum_j \sum_i D_{im}\, y_{ijm} \ln \frac{\sigma_{jm}^{z_{im}}(\theta)}{\bar{\sigma}_{jm}(\theta)}}_{\text{micro}} + \underbrace{\sum_m N_m \sum_j s_{jm} \ln \bar{\sigma}_{jm}(\theta)}_{\text{macro}},$$

where $D_{im} = 1$ if i is in the micro sample, $\sigma_{jm}^{z_{im}}$ is the choice probability conditional on demographics z_{im}, and $\bar{\sigma}_{jm}$ is the unconditional choice probability.

Open Question: In the Census/LEHD application with near-complete coverage, do we treat $I_m \approx N_m$ (so the macro likelihood term vanishes) or maintain a micro/macro split?

2.2 Macro Component: Firm-Level Moment Conditions

The macro component uses four firm-level population moment conditions to identify $(\eta, \alpha, \sigma_e, \gamma_0)$. We present the moments first for Specification M, then for Specification A. We then describe the computation of average skill and write the sample analogs.

Throughout, we use the standardized objects from the micro estimation: $\tilde{q}_j$, $\tilde{\gamma}$, $\{\delta_j\}$, and τ. Define standardized log skill $\tilde{s}_i \equiv \tilde{\gamma}\, v_i + \tilde{e}_i$ where $\tilde{e}_i \sim N(0, 1)$.

2.2.1 Screening FOC in Standardized Form

The screening FOC $w_j = \bar{q}_j \cdot A_j(1-\alpha)(Q_j L_j)^{-\alpha}$ can be written in logs. Under Specification M, $\ln \bar{q}_j = \gamma_0 + \sigma_e \tilde{q}_j$ and $\ln Q_j = \gamma_0 + \ln \tilde{Q}_j^M(\sigma_e)$, giving

$$\underbrace{\ln(1-\alpha) + (1-\alpha)\gamma_0 + \sigma_e \tilde{q}_j - \alpha \ln \tilde{Q}_j^M(\sigma_e) - \alpha \ln L_j - \ln w_j}_{= -\ln A_j} = 0 \quad (\text{up to } \ln A_j). \tag{5}$$

Under Specification A, $\ln \bar{q}_j = \ln(\gamma_0 + \sigma_e \tilde{q}_j)$ and $\ln Q_j = \ln(\gamma_0 + \sigma_e \tilde{Q}_j^A)$, giving

$$\underbrace{\ln(1-\alpha) + \ln(\gamma_0 + \sigma_e \tilde{q}_j) - \alpha \ln(\gamma_0 + \sigma_e \tilde{Q}_j^A) - \alpha \ln L_j - \ln w_j}_{= -\ln A_j} = 0 \quad (\text{up to } \ln A_j). \tag{6}$$

Additionally, combining the screening FOC with $R_j = A_j H_j^{1-\alpha}$ eliminates TFP:

$$\ln \bar{q}_j - \ln Q_j = \ln w_j + \ln L_j - \ln(1-\alpha) - \ln R_j. \tag{7}$$

2.2.2 Population Moments: Specification M

Moment 1: Preference (identifies η).

$$\mathbb{E}\Big[\delta_j - \eta \ln w_j \;\Big|\; z_j^1\Big] = 0, \tag{8}$$

where z_j^1 is an exogenous shock to TFP (which shifts labor demand, hence wages, without directly affecting amenities $\xi_j = \delta_j - \eta \ln w_j$). This is the standard BLP-type demand inversion moment.

Moment 2: Screening FOC (identifies α).

$$\mathbb{E}\Big[\ln(1-\alpha) + (1-\alpha)\gamma_0 + \sigma_e \tilde{q}_j - \alpha \ln \tilde{Q}_j^M(\sigma_e) - \alpha \ln L_j - \ln w_j \;\Big|\; z_j^2\Big] = 0, \tag{9}$$

where z_j^2 is an exogenous shock to amenities (which shifts labor supply, hence wages and workforce composition, without directly affecting TFP). The residual is $-\ln A_j$, and α is the coefficient on $\ln(\tilde{Q}_j^M \cdot L_j)$. This moment requires $\mathbb{E}[\ln A_j \mid z_j^2] = 0$.

Moment 3: A_j-free / revenue control (identifies σ_e). Substituting (**??**) in standardized form:

$$\mathbb{E}\Big[\sigma_e \tilde{q}_j - \ln \tilde{Q}_j^M(\sigma_e) - \ln w_j - \ln L_j + \ln(1-\alpha) + \ln R_j\Big] = 0. \tag{10}$$

Revenue acts as a control function that eliminates A_j entirely from this moment—there is no structural residual. At the true parameters, the expression inside the expectation equals zero for every firm j, not merely in expectation. As a result, any function of observables can be used to form additional moment conditions; there is no exogeneity requirement because there is nothing to be orthogonal to. Formally, this is analogous to an independent measurement error model: if one treats any discrepancy from zero as classical measurement error (independent of all observables), then any variable is a valid instrument. Under Spec M, γ_0 cancels (it enters $\ln \bar{q}_j$ and $\ln Q_j$ symmetrically), so a single moment ($z_j = 1$) suffices for the single unknown σ_e. Additional interactions (e.g., $z_j = \tilde{q}_j$) provide overidentification.

Different values of σ_e generate different gaps between $\sigma_e \tilde{q}_j$ and $\ln \tilde{Q}_j^M(\sigma_e)$ across firms—the convexity of $\exp(\sigma_e \cdot)$ inflates average skill relative to marginal skill more at high-threshold firms when σ_e is large.

Moment 4: TFP normalization (identifies γ_0). Taking the unconditional mean of (**??**) and imposing $\mathbb{E}[\ln A_j] = 0$:

$$\mathbb{E}\Big[\ln(1-\alpha) + (1-\alpha)\gamma_0 + \sigma_e \tilde{q}_j - \alpha \ln \tilde{Q}_j^M(\sigma_e) - \alpha \ln L_j - \ln w_j\Big] = 0. \tag{11}$$

This is Moment 2 evaluated at $z_j^2 = 1$. Given (α, σ_e) from Moments 2–3, it pins γ_0 in closed form:

$$\gamma_0 = \frac{1}{1-\alpha}\left[\mathbb{E}\Big[\ln w_j + \alpha \ln \tilde{Q}_j^M(\sigma_e) + \alpha \ln L_j\Big] - \mathbb{E}[\sigma_e \tilde{q}_j] - \ln(1-\alpha)\right]. \tag{12}$$

Equivalently, using revenue: $\gamma_0 = \frac{1}{1-\alpha}\left[\mathbb{E}[\ln R_j] - (1-\alpha)\mathbb{E}[\ln(\tilde{Q}_j^M(\sigma_e) \cdot L_j)]\right]$. The interpretation: γ_0 is the level of log human capital needed to rationalize observed revenues given normalized TFP.

Why γ_0 requires a TFP normalization under Spec M. Shifting γ_0 by c multiplies all q_i by e^c, which is absorbed by $A_j \to A_j e^{-c(1-\alpha)}$. The screening FOC residual $-\ln A_j$ shifts by $(1-\alpha)c$ uniformly across firms. Moment 2 (with demeaned instruments z_j^2) is insensitive to this shift; only the level restriction in Moment 4 breaks the degeneracy.

2.2.3 Population Moments: Specification A

Moments 1 and 2 have the same structure, with $\ln \bar{q}_j$ and $\ln Q_j$ replaced by their Spec A expressions.

Moment 1: Preference (identifies η). Identical to (**??**):

$$\mathbb{E}\Big[\delta_j - \eta \ln w_j \;\Big|\; z_j^1\Big] = 0. \tag{13}$$

Moment 2: Screening FOC (identifies α).

$$\mathbb{E}\Big[\ln(1-\alpha) + \ln(\gamma_0 + \sigma_e \tilde{q}_j) - \alpha \ln(\gamma_0 + \sigma_e \tilde{Q}_j^A) - \alpha \ln L_j - \ln w_j \;\Big|\; z_j^2\Big] = 0. \tag{14}$$

Moment 3: A_j-free / revenue control (identifies σ_e and γ_0).

$$\mathbb{E}\Big[\ln(\gamma_0 + \sigma_e \tilde{q}_j) - \ln(\gamma_0 + \sigma_e \tilde{Q}_j^A) - \ln w_j - \ln L_j + \ln(1-\alpha) + \ln R_j\Big] = 0. \tag{15}$$

As in Spec M, revenue eliminates A_j entirely—there is no structural residual, so any function of observables can be used to form moments (see the discussion under Moment 3 for Spec M). Unlike Spec M, γ_0 does *not* cancel: $\ln(\gamma_0 + \sigma_e \tilde{q}_j) - \ln(\gamma_0 + \sigma_e \tilde{Q}_j^A) \neq$ const as a function of γ_0, because the additive structure breaks scale invariance. Two moments (e.g., interacted with $z_j = 1$ and $z_j = \tilde{q}_j$) suffice to identify both γ_0 and σ_e from this moment alone, with no TFP assumption.

Moment 4: TFP normalization (overidentifying under Spec A).

$$\mathbb{E}\Big[\ln(1-\alpha) + \ln(\gamma_0 + \sigma_e \tilde{q}_j) - \alpha \ln(\gamma_0 + \sigma_e \tilde{Q}_j^A) - \alpha \ln L_j - \ln w_j\Big] = 0. \tag{16}$$

Under Spec A, since Moment 3 already identifies both (γ_0, σ_e), Moment 4 is available as an overidentifying restriction (or can be dropped). Under Spec M, it is essential for γ_0.

2.2.4 Summary

Moment	Identifies	Spec. A	Spec. M
1: $\mathbb{E}[\xi_j \mid z_j^1] = 0$	η	same	same
2: $\mathbb{E}[\ln A_j \mid z_j^2] = 0$	α	(**??**)	(**??**)
3: A_j-free (revenue)	σ_e (+ γ_0 in A)	(**??**)	(**??**)
4: $\mathbb{E}[\ln A_j] = 0$	γ_0	overidentifying	essential

2.2.5 Computing Average Skill

The moments above require model-implied average skill $\tilde{Q}_j$. For each matched worker i with $y_i = j$, we need the posterior expectation of a function of $\tilde{e}_i$ given the observed match. Using the ordered-threshold structure, define $a_{ik} \equiv \tilde{q}_{o^{-1}(k)} - \tilde{\gamma}\, v_i$ and $b_{ik} \equiv \tilde{q}_{o^{-1}(k+1)} - \tilde{\gamma}\, v_i$, and let $\pi_{jk}(d_i) \equiv \frac{\exp(\delta_j - \tau d_{ij})}{1+\sum_{m=1}^{k} \exp(\delta_{o^{-1}(m)} - \tau d_{i,o^{-1}(m)})}$ be the logit probability of choosing j given choice set $O_k = \{o^{-1}(1), \ldots, o^{-1}(k)\}$. Then for a generic function $h(\tilde{e}_i)$:

$$\mathbb{E}\Big[h(\tilde{e}_i) \,\Big|\, y_i = j, v_i, d_i\Big] = \frac{1}{\sigma_j(v_i, d_i)} \sum_{k=0}^{J} \mathbf{1}[\tilde{q}_j \le \tilde{q}_{o^{-1}(k)}]\, \pi_{jk}(d_i)\ \left(\Phi(b_{ik}) - \Phi(a_{ik})\right)\ \mathbb{E}\Big[h(\tilde{e}_i) \,\Big|\, \tilde{e}_i \in [a_{ik},\, b_{ik}]\Big]. \tag{17}$$

Specification A. Set $h(\tilde{e}) = \tilde{e}$. With $\tilde{e}_i \sim N(0,1)$, the truncated expectation has the closed form

$$\mathbb{E}[\tilde{e}_i \mid \tilde{e}_i \in [a,b]] = \frac{\phi(a) - \phi(b)}{\Phi(b) - \Phi(a)}.$$

Standardized average skill at firm j is

$$\tilde{Q}_j^A = \frac{1}{L_j} \sum_{i:\, y_i = j} \left(\tilde{\gamma}\, v_i + \mathbb{E}[\tilde{e}_i \mid y_i = j, v_i, d_i]\right). \tag{18}$$

This quantity is fully determined by θ_{micro} and does not depend on σ_e. Structural average skill is $Q_j = \gamma_0 + \sigma_e \tilde{Q}_j^A$.

Specification M. Set $h(\tilde{e}) = \exp(\sigma_e \tilde{e})$. The truncated expectation is

$$\mathbb{E}[\exp(\sigma_e \tilde{e}_i) \mid \tilde{e}_i \in [a,b]] = \exp\left(\tfrac{\sigma_e^2}{2}\right) \frac{\Phi(b - \sigma_e) - \Phi(a - \sigma_e)}{\Phi(b) - \Phi(a)}.$$

Standardized average skill at firm j is

$$\tilde{Q}_j^M(\sigma_e) = \frac{1}{L_j} \sum_{i:\, y_i = j} \exp(\sigma_e \tilde{\gamma}\, v_i)\ \mathbb{E}[\exp(\sigma_e \tilde{e}_i) \mid y_i = j, v_i, d_i]. \tag{19}$$

This quantity depends on σ_e even though all other inputs come from micro estimation: the convexity of $\exp(\sigma_e \cdot)$ means that different values of σ_e imply different within-firm skill dispersions in level units. Structural average skill is $Q_j = e^{\gamma_0} \tilde{Q}_j^M(\sigma_e)$.

2.2.6 Sample Moment Conditions

Write $\hat{\mathbb{E}}_j[\cdot] \equiv \frac{1}{J} \sum_{j=1}^{J} (\cdot)$ for the cross-firm average. The sample analogs of Moments 1–4 under Specification M are:

Moment 1 (preference):

$$\hat{m}_1(\theta) = \frac{1}{J}\sum_{j=1}^{J} z_j^1 \cdot (\delta_j - \eta \ln w_j). \tag{20}$$

Moment 2 (screening FOC):

$$\hat{m}_2^M(\theta) = \frac{1}{J}\sum_{j=1}^{J} z_j^2 \cdot \Big[\ln(1-\alpha) + (1-\alpha)\gamma_0 + \sigma_e \tilde{q}_j - \alpha \ln \hat{\tilde{Q}}_j^M(\sigma_e) - \alpha \ln L_j - \ln w_j\Big]. \tag{21}$$

Moment 3 (A_j-free): Since revenue eliminates A_j, there is no structural residual and any function of observables can serve as an interaction variable (formally, any discrepancy is independent measurement error). Under Spec M, where only σ_e is unknown, $z_j = 1$ suffices:

$$\hat{m}_3^M(\theta) = \frac{1}{J}\sum_{j=1}^{J} \Big[\sigma_e \tilde{q}_j - \ln \hat{\tilde{Q}}_j^M(\sigma_e) - \ln w_j - \ln L_j + \ln(1-\alpha) + \ln R_j\Big]. \tag{22}$$

Moment 4 (TFP normalization):

$$\hat{m}_4^M(\theta) = \frac{1}{J}\sum_{j=1}^{J} \Big[\ln(1-\alpha) + (1-\alpha)\gamma_0 + \sigma_e \tilde{q}_j - \alpha \ln \hat{\tilde{Q}}_j^M(\sigma_e) - \alpha \ln L_j - \ln w_j\Big]. \tag{23}$$

Under Specification A, replace $\sigma_e \tilde{q}_j$ with $\ln(\gamma_0 + \sigma_e \tilde{q}_j)$ and $\ln \tilde{Q}_j^M(\sigma_e)$ with $\ln(\gamma_0 + \sigma_e \hat{\tilde{Q}}_j^A)$ throughout Moments 2–4, and correspondingly in Moment 3. Explicitly:

Moment 2 (Spec A):

$$\hat{m}_2^A(\theta) = \frac{1}{J}\sum_{j=1}^{J} z_j^2 \cdot \Big[\ln(1-\alpha) + \ln(\gamma_0 + \sigma_e \tilde{q}_j) - \alpha \ln(\gamma_0 + \sigma_e \hat{\tilde{Q}}_j^A) - \alpha \ln L_j - \ln w_j\Big]. \tag{24}$$

Moment 3 (Spec A): Under Spec A, both γ_0 and σ_e are unknown, so we need two moment conditions. Since there is no structural residual, we are free to choose any interactions; $z_j \in \{1, \tilde{q}_j\}$ is a natural choice:

$$\hat{m}_3^A(\theta) = \frac{1}{J}\sum_{j=1}^{J} \begin{pmatrix} 1 \\ \tilde{q}_j \end{pmatrix} \cdot \Big[\ln(\gamma_0 + \sigma_e \tilde{q}_j) - \ln(\gamma_0 + \sigma_e \hat{\tilde{Q}}_j^A) - \ln w_j - \ln L_j + \ln(1-\alpha) + \ln R_j\Big]. \tag{25}$$

Moment 4 (Spec A):

$$\hat{m}_4^A(\theta) = \frac{1}{J}\sum_{j=1}^{J} \Big[\ln(1-\alpha) + \ln(\gamma_0 + \sigma_e \tilde{q}_j) - \alpha \ln(\gamma_0 + \sigma_e \hat{\tilde{Q}}_j^A) - \alpha \ln L_j - \ln w_j\Big]. \tag{26}$$

In both specifications, $\hat{\tilde{Q}}_j$ is computed from (**??**) or (**??**) using the micro-estimated θ_{micro} and the matched workers at firm j.

The stacked moment vector is

$$\hat{m}(\theta) = \begin{pmatrix} \hat{m}_1(\theta) \\ \hat{m}_2(\theta) \\ \hat{m}_3(\theta) \\ \hat{m}_4(\theta) \end{pmatrix}, \tag{27}$$

with $\hat{m}_2$–$\hat{m}_4$ chosen according to the specification. The weighting matrix $\hat{W}$ is positive definite; an initial choice is block-diagonal with blocks corresponding to each moment group, or a 2-step optimal GMM weighting matrix.

3 Estimation Variants

Four variants along two dimensions: (1) MLE vs. GMM for the micro component, and (2) whether to search over $(\tilde{q}, \delta)$ jointly or concentrate out δ. Note that (γ_0, σ_e) enter only through the macro penalty $\hat{m}(\theta)$, not through the micro likelihood; they can be concentrated out or searched over separately.

3.1 Variant 1: MLE – Search over $(\tilde{q}, \delta)$

$$\min_{\tilde{\gamma},\tau,\gamma_0,\sigma_e,\alpha,\eta,\tilde{q},\delta} -\sum_i \ln \sigma_{y_i}(v_i, d_i; \tilde{\gamma}, \tau, \tilde{q}, \delta) \; + \; \frac{1}{2}\hat{m}(\tilde{\gamma}, \tau, \gamma_0, \sigma_e, \alpha, \eta, \tilde{q}, \delta)'\hat{W}\,\hat{m}(\cdot) \tag{28}$$

3.2 Variant 2: MLE – Concentrate out δ

$$\min_{\tilde{\gamma},\tau,\gamma_0,\sigma_e,\alpha,\eta,\tilde{q}} -\sum_i \ln \sigma_{y_i}(v_i, d_i; \tilde{\gamma}, \tau, \tilde{q}, \hat{\delta}(\tilde{\gamma}, \tau, \tilde{q})) \; + \; \frac{1}{2}\hat{m}(\cdot)'\hat{W}\,\hat{m}(\cdot)$$

where $\hat{\delta}(\tilde{\gamma}, \tau, \tilde{q})$ solves the share-matching fixed point:

$$\frac{1}{N}\sum_i \mathbf{1}[y_i = j] = \frac{1}{N}\sum_i \sigma_j(v_i, d_i; \tilde{\gamma}, \tau, \tilde{q}, \hat{\delta}) \quad \forall\, j = 1, \ldots, J, \tag{29}$$

with the BLP-style contraction mapping:

$$\hat{\delta}_j^{(p+1)} \leftarrow \hat{\delta}_j^{(p)} + \ln\left(\frac{1}{N}\sum_i \mathbf{1}[y_i = j]\right) - \ln\left(\frac{1}{N}\sum_i \sigma_j(v_i, d_i; \tilde{\gamma}, \tau, \tilde{q}, \hat{\delta}^{(p)})\right).$$

Open Question: Does the contraction mapping remain valid with screening? Conditional on fixed $\tilde{q}$, the choice set structure is held fixed and σ_j is monotone in δ_j within each choice set, so we believe it works. Needs verification.

3.3 Variant 3: GMM – Search over $(\tilde{q}, \delta)$

$$\min_{\tilde{\gamma},\tau,\gamma_0,\sigma_e,\alpha,\eta,\tilde{q},\delta} \left\| \sum_i \sum_{j=1}^{J} (y_{ij} - \sigma_j(v_i, d_i; \theta_{\text{micro}}))\, \nabla_{\theta_{\text{micro}}} \ln \frac{\sigma_j(v_i, d_i; \theta_{\text{micro}}^{(1)})}{\sigma_0(v_i, d_i; \theta_{\text{micro}}^{(1)})} \right\|^2_{\hat{W}_\mu} \; + \; \frac{1}{2}\hat{m}(\theta)'\hat{W}\,\hat{m}(\theta) \tag{30}$$

where $\theta_{\text{micro}}^{(1)}$ is a first-step estimate.

3.4 Variant 4: GMM – Concentrate out δ

$$\min_{\tilde{\gamma},\tau,\gamma_0,\sigma_e,\alpha,\eta,\tilde{q}} \left\| \sum_i \sum_{j=1}^{J} \left(y_{ij} - \sigma_j(v_i, d_i; \tilde{\gamma}, \tau, \tilde{q}, \hat{\delta}) \right) \nabla_{\theta_{\text{micro}}} \ln \frac{\sigma_j(\cdot; \theta^{(1)}_{\text{micro}})}{\sigma_0(\cdot; \theta^{(1)}_{\text{micro}})} \right\|^2_{\hat{W}_\mu} + \frac{1}{2} \hat{m}(\cdot)' \hat{W} \, \hat{m}(\cdot) \quad (31)$$

Open Question: We do not consider concentrating out $\tilde{q}$. Inverting for $\tilde{q}$ is less standard than the BLP δ-inversion and the ordering of firms by thresholds could switch, causing discontinuities.

Comparison. Based on simulations (50 firms, 1000 workers): MLE is faster ($\sim$3s vs. $\sim$60s) and appears less biased. The share-inversion variants are attractive with multiple markets (parallel inversion by market). The standardized reparametrization has the additional advantage that the micro likelihood depends on fewer parameters $(\tilde{\gamma}, \tau, \{\tilde{q}_j, \delta_j\})$, with $(\gamma_0, \sigma_e, \alpha, \eta)$ entering only through the macro penalty.

4 Initialization

Step 0a: Skill parameters from wages. Pool all matched workers across markets. Regress $\ln w_i$ on $(1, v_i)$: $\gamma_0^0 = \hat{\beta}_0$, $\sigma_e^0 = \sqrt{\text{Var}(\hat{\varepsilon})}$, $\tilde{\gamma}^0 = \hat{\beta}_1/\sigma_e^0$.

Step 0b: Firm average skill. For each firm j in market m, compute $\hat{Q}_j = \frac{1}{L_j} \sum_{i:y_i=j} \exp(\gamma_0^0 + \sigma_e^0 \tilde{\gamma}^0 v_i)$.

Step 0c: Production function (2SLS). Pool all firms. Regress $\ln R_j$ on $\ln(\hat{Q}_j L_j)$ instrumented with $(1, z_j^2)$. The coefficient estimates $(1-\alpha)$; set $\alpha^0 = 1 - \hat{\beta}$. Clamp to $(0, 1)$ or default 0.25.

Step 0d: Screening thresholds. From the screening FOC: $\ln \bar{q}_j^0 = \ln w_j + \ln \hat{Q}_j + \ln L_j - \ln R_j - \ln(1-\alpha^0)$. Standardize: $\tilde{q}_j^0 = (\ln \bar{q}_j^0 - \gamma_0^0)/\sigma_e^0$.

Step 0e: Naive preference parameters. Standard logit ignoring screening (all firms in every choice set) per market. Pool τ as N-weighted average across markets. This identifies τ^0 and δ_j^0.

Step 0f: Wage equation (2SLS). Pool all firms. Regress δ_j^0 on $\ln w_j$ instrumented with $(1, z_j^1)$. The coefficient is η^0.

5 Weighting Matrix

Initial choice. $\hat{W} = (\bar{B}'\bar{B})^{-1}$ where $\bar{B}$ is the $\bar{J} \times d_b$ matrix with rows b'_{jm}.

Two-step optimal GMM. (1) Estimate $\theta^{(1)}$ with initial $\hat{W}$. (2) Construct $\hat{W}^{\text{opt}}$ from residuals. (3) Re-estimate.

Scaling. The $\frac{1}{2}$ in the GMM penalty (Grieco et al. eq. (9)) affects relative weight on $\log \hat{L}$ vs. $\hat{\chi}$ and the conformance property.

Open Question: Single weighting matrix $\hat{W}$ for stacked moments, or block-diagonal (separate weights for preference and screening moments)?

Open Question: For the GMM interpretation (Variant 3), Grieco et al. show asymptotic equivalence requires $\hat{W}_L = (-\partial_{\psi\psi'} \log \hat{L})^{-1}$. Feasible with $2J$ fixed effects?

6 Multi-Market Extension

With M markets: $\theta = (\theta_G, (\theta_m)_{m=1}^M)$, $\theta_G = (\tilde{\gamma}, \tau, \gamma_0, \sigma_e, \alpha, \eta)$, $\theta_m = (\delta_{j_m}, \tilde{q}_{j_m})_{j_m}$. Note that the micro likelihood for market m depends on θ_m and $(\tilde{\gamma}, \tau)$ only; the remaining global parameters $(\gamma_0, \sigma_e, \alpha, \eta)$ enter only through the macro penalty.

6.1 MLE: Distributed Optimization

The likelihood decomposes: $-\ell(\theta) = -\sum_m \ell_m(\theta_G, \theta_m)$. Profile out θ_m:

$$\min_{\theta_G} \sum_{m=1}^{M} \min_{\theta_m} -\ell_m(\theta_G, \theta_m).$$

By the envelope theorem: $\nabla_{\theta_G} \tilde{\ell}(\theta_G) = \sum_m \partial_{\theta_G} \ell_m \big|_{\theta_m = \hat{\theta}_m(\theta_G)}$.

The profiled Hessian (Schur complement) is

$$\tilde{H}_{GG} = \sum_m \left(\nabla^2_{\theta_G \theta_G'} \ell_m - (\nabla^2_{\theta_G \theta_m'} \ell_m)(\nabla^2_{\theta_m \theta_m'} \ell_m)^{-1} (\nabla^2_{\theta_m \theta_G'} \ell_m) \right).$$

Per-step communication. Each market-m worker: (1) receives $\theta_G^{(t)}$; (2) solves inner problem for $\theta_m^{(t)}$; (3) returns $\ell_m^{(t)}$, $\nabla_{\theta_G} \ell_m$, and Hessian blocks.

6.2 Augmented-Lagrangian Formulation for the Hybrid Objective

The pure MLE decomposition above ignores the GMM penalty. For the full hybrid objective, the difficulty is that the MLE term is additive across markets but the quadratic GMM penalty is not. The augmented-Lagrangian reformulation below solves the exact hybrid objective in a distributed fashion, with no approximations to the target estimator. All approximations are purely numerical (finite solver tolerances, finite outer iterations).

6.2.1 Setup and Notation

Recall the parameter partition: $(\tau, \tilde{\gamma})$ enter the MLE; $(\alpha, \sigma_e, \eta, \gamma_0, \tilde{\gamma})$ enter the GMM; $\tilde{\gamma}$ is shared. Market m's local parameters are $\theta_m = (\delta_m, \tilde{q}_m)$. Let D_m denote the microdata in market m, N_m the number of firms, and define market weights

$$\omega_m = \frac{N_m}{\sum_{j=1}^M N_j}.$$

The market-level log-likelihood contribution is $\ell_m(\tau, \tilde{\gamma}, \delta_m, \tilde{q}_m; D_m)$ and the market-level moment contribution is $g_m(\alpha, \sigma_e, \eta, \gamma_0, \tilde{\gamma}, \delta_m, \tilde{q}_m; D_m)$, stacking the sample analogs of Moments 1–4 from Section **??**. The aggregate moment is $\bar{g} = \sum_{m=1}^{M} \omega_m g_m$.

The exact hybrid objective is

$$Q = \sum_{m=1}^{M} \ell_m(\tau, \tilde{\gamma}, \delta_m, \tilde{q}_m; D_m) + \frac{1}{2}\bar{g}' \, W \, \bar{g}. \tag{32}$$

The distributed algorithm below targets a stationary point of exactly this objective.

6.2.2 Exact Constrained Reformulation

Introduce an auxiliary variable μ satisfying $\mu = \bar{g}$. Then (**??**) is equivalent to

$$\min_{\tau, \tilde{\gamma}, \alpha, \sigma_e, \eta, \gamma_0, \{\delta_m, \tilde{q}_m\}, \mu} \sum_{m=1}^{M} \ell_m(\tau, \tilde{\gamma}, \delta_m, \tilde{q}_m; D_m) + \frac{1}{2}\mu' W \mu \tag{33}$$

subject to $\mu - \sum_{m=1}^{M} \omega_m g_m(\alpha, \sigma_e, \eta, \gamma_0, \tilde{\gamma}, \delta_m, \tilde{q}_m; D_m) = 0$. This is a re-expression, not an approximation.

6.2.3 Augmented Lagrangian

Let ν be the multiplier and $\rho > 0$ the penalty parameter. The augmented Lagrangian is

$$\begin{aligned} \mathcal{A}_\rho = & \sum_{m=1}^{M} \ell_m(\tau, \tilde{\gamma}, \delta_m, \tilde{q}_m; D_m) + \frac{1}{2}\mu' W \mu \\ & + \nu' \left(\mu - \sum_{m=1}^{M} \omega_m g_m \right) + \frac{\rho}{2} \left\| \mu - \sum_{m=1}^{M} \omega_m g_m \right\|^2 . \end{aligned} \tag{34}$$

Its role is purely computational. At convergence the constraint $\mu = \bar{g}$ holds, so a KKT point of (**??**) corresponds to a stationary point of the original hybrid objective (**??**).

6.2.4 One Outer Iteration

Suppose the current values at iteration k are $(\tau^k, \tilde{\gamma}^k, \alpha^k, \sigma_e^k, \eta^k, \gamma_0^k)$, $(\delta_m^k, \tilde{q}_m^k)_{m=1}^{M}$, μ^k, ν^k, ρ_k. Define the current market moments $g_m^k = g_m(\alpha^k, \sigma_e^k, \eta^k, \gamma_0^k, \tilde{\gamma}^k, \delta_m^k, \tilde{q}_m^k; D_m)$ and the aggregate $\bar{g}^k = \sum_m \omega_m g_m^k$.

Step 1: Broadcast. The coordinator sends to each worker m:

$$(\tau^k, \tilde{\gamma}^k, \alpha^k, \sigma_e^k, \eta^k, \gamma_0^k, \nu^k, \rho_k),$$

and a market-specific residual target

$$c_m^k = \mu^k - \bar{g}^k + \omega_m g_m^k. \tag{35}$$

This can be computed from the previously cached $\bar{g}^k$ and g_m^k without accessing microdata.

Step 2: Market block update (parallel). Holding all global and other-market variables fixed, worker m solves the exact block subproblem

$$(\delta_m^{k+1}, \tilde{q}_m^{k+1}) \approx \arg\min_{\delta_m, \tilde{q}_m} \Bigg\{ \ell_m(\tau^k, \tilde{\gamma}^k, \delta_m, \tilde{q}_m; D_m) - \omega_m \nu^{k\prime} g_m(\alpha^k, \sigma_e^k, \eta^k, \gamma_0^k, \tilde{\gamma}^k, \delta_m, \tilde{q}_m; D_m) + \frac{\rho_k}{2} \left\| c_m^k - \omega_m g_m(\alpha^k, \sigma_e^k, \eta^k, \gamma_0^k, \tilde{\gamma}^k, \delta_m, \tilde{q}_m; D_m) \right\|^2 \Bigg\}. \tag{36}$$

This is the exact market block of the augmented Lagrangian. Each worker uses only its own microdata D_m.

Step 2 return values. After solving, worker m returns:

(i) δ_m^{k+1}, $\tilde{q}_m^{k+1}$;

(ii) $g_m^{k+1,k} = g_m(\alpha^k, \sigma_e^k, \eta^k, \gamma_0^k, \tilde{\gamma}^k, \delta_m^{k+1}, \tilde{q}_m^{k+1}; D_m)$;

(iii) $\ell_m^{k+1,k} = \ell_m(\tau^k, \tilde{\gamma}^k, \delta_m^{k+1}, \tilde{q}_m^{k+1}; D_m)$.

Step 3: Intermediate μ update (closed form). The coordinator aggregates $\bar{g}^{k+1,k} = \sum_m \omega_m g_m^{k+1,k}$ and updates

$$\mu^{k+\frac{1}{2}} = (W + \rho_k I)^{-1}(\rho_k \bar{g}^{k+1,k} - \nu^k). \tag{37}$$

Step 4: Global block update (callback-based). Holding $(\delta_m^{k+1}, \tilde{q}_m^{k+1})$ fixed for all m, solve the concentrated global problem

$$(\tau^{k+1}, \tilde{\gamma}^{k+1}, \alpha^{k+1}, \sigma_e^{k+1}, \eta^{k+1}, \gamma_0^{k+1}) \approx \arg\min_{\tau, \tilde{\gamma}, \alpha, \sigma_e, \eta, \gamma_0} \Phi_k(\tau, \tilde{\gamma}, \alpha, \sigma_e, \eta, \gamma_0), \tag{38}$$

where

$$\Phi_k = \sum_{m=1}^{M} \ell_m(\tau, \tilde{\gamma}, \delta_m^{k+1}, \tilde{q}_m^{k+1}; D_m) + \frac{1}{2}\mu^{*\prime} W \mu^* + \nu^{k\prime}(\mu^* - \bar{g}) + \frac{\rho_k}{2}\|\mu^* - \bar{g}\|^2, \tag{39}$$

with $\bar{g} = \sum_m \omega_m g_m(\alpha, \sigma_e, \eta, \gamma_0, \tilde{\gamma}, \delta_m^{k+1}, \tilde{q}_m^{k+1}; D_m)$ and $\mu^* = (W + \rho_k I)^{-1}(\rho_k \bar{g} - \nu^k)$ concentrated out analytically.

This is a 6-dimensional unconstrained problem. The coordinator hands it to an off-the-shelf solver (Section **??**). Each function/gradient evaluation by the solver triggers a *worker callback* (Section **??**).

Step 5: Final μ and multiplier update. After the global solve, each worker evaluates g_m^{k+1} and ℓ_m^{k+1} at the accepted global point (one final callback). The coordinator aggregates $\bar{g}^{k+1} = \sum_m \omega_m g_m^{k+1}$ and updates

$$\mu^{k+1} = (W + \rho_k I)^{-1}(\rho_k \bar{g}^{k+1} - \nu^k), \tag{40}$$

$$\nu^{k+1} = \nu^k + \rho_k(\mu^{k+1} - \bar{g}^{k+1}). \tag{41}$$

Step 6: Penalty update and convergence check. Update ρ_k (Section **??**). Stop when the primal residual $\|\mu^{k+1} - \bar{g}^{k+1}\|$, dual residual $\rho_k\|\bar{g}^{k+1} - \bar{g}^k\|$, and parameter changes are all small (Section **??**).

6.2.5 Worker Callbacks During the Global Solve

For each trial global point $(\tau, \tilde{\gamma}, \alpha, \sigma_e, \eta, \gamma_0)$ proposed by the coordinator's solver:

1. The coordinator broadcasts the trial point to all workers.
2. Each worker m holds $(\delta_m^{k+1}, \tilde{q}_m^{k+1})$ fixed and computes, using its own microdata:

$$L_m = \ell_m(\tau, \tilde{\gamma}, \delta_m^{k+1}, \tilde{q}_m^{k+1}; D_m),$$
$$G_m = g_m(\alpha, \sigma_e, \eta, \gamma_0, \tilde{\gamma}, \delta_m^{k+1}, \tilde{q}_m^{k+1}; D_m).$$

   If the solver requires gradients, the worker also computes:

$$S_m = \nabla_{(\tau,\tilde{\gamma})}\ell_m\big|_{\delta_m^{k+1},\tilde{q}_m^{k+1}},$$
$$J_m = \nabla_{(\alpha,\sigma_e,\eta,\gamma_0,\tilde{\gamma})}g_m\big|_{\delta_m^{k+1},\tilde{q}_m^{k+1}}.$$

3. Worker m returns (L_m, G_m, S_m, J_m) to the coordinator.
4. The coordinator aggregates:

$$L = \sum_m L_m, \quad \bar{g} = \sum_m \omega_m G_m, \quad S = \sum_m S_m, \quad \bar{J} = \sum_m \omega_m J_m,$$

   computes $\mu^* = (W + \rho_k I)^{-1}(\rho_k\bar{g} - \nu^k)$, evaluates Φ_k and its gradient, and returns these to the solver.

This design has two key properties. First, every function/gradient evaluation is *exact* — there are no quadratic approximations to the likelihood or moment functions. The solver sees exact values and gradients of Φ_k, so it can converge to an accurate solution of the global subproblem. Second, no machine ever holds more than one market's microdata. The coordinator works exclusively with the low-dimensional aggregates $(L, \bar{g}, S, \bar{J})$.

Why all parameters can be updated exactly. In the previous architecture, parameters like σ_e under Spec M required worker-level computation because $\tilde{Q}_j^M(\sigma_e)$ involves $E[\exp(\sigma_e\tilde{e}_i) \mid \tilde{e}_i \in [a_{ik}, b_{ik}]]$, which is not separable from the worker-specific truncation bounds (a_{ik}, b_{ik}). The callback design avoids this problem entirely: the worker evaluates g_m at the trial σ_e by iterating over its matched workers (a forward pass, not an optimization), and returns only the low-dimensional result. The coordinator never needs to know the truncation bounds.

Why market derivatives are not missing. When the coordinator solves for the global parameters, the local parameters $(\delta_m^{k+1}, \tilde{q}_m^{k+1})$ are held fixed. The coordinator differentiates only with respect to $(\tau, \tilde{\gamma}, \alpha, \sigma_e, \eta, \gamma_0)$. The derivatives with respect to market-specific parameters were already handled in the market block solve (Step 2).

6.2.6 What Is Exact and What Is Approximate

The following are exact: the original hybrid objective (**??**); the constrained reformulation; the augmented Lagrangian (**??**); each block subproblem conditional on the other blocks; the closed-form update for μ; the multiplier update for ν; and every function/gradient evaluation in the worker callbacks.

The following are numerical approximations (inherent to any iterative algorithm, not modifications of the estimator): solving each block subproblem to finite tolerance; using a finite number of outer iterations; the heuristic residual-balancing rule for ρ.

6.2.7 Implementation Specifications

Market subproblem solver. Solve (**??**) using L-BFGS via `scipy.optimize.minimize(method='L-BFGS-B')` or `jaxopt.LBFGS`, searching jointly over $(\delta_m, \tilde{q}_m)$. Warm-start from $(\delta_m^k, \tilde{q}_m^k)$. Gradients are computed by JAX reverse-mode autodiff through the combined objective (likelihood plus penalty terms). The BLP contraction for δ_m can optionally be used to get a good starting point before switching to L-BFGS for the joint problem including the penalty terms.

Convergence tolerance: gradient norm $\leq 10^{-6}$. Maximum inner iterations: 200. If the subproblem fails to converge, return the best iterate found and flag it; the outer loop is robust to moderately inexact block solves.

Global solver. Solve (**??**) using `scipy.optimize.minimize(method='L-BFGS-B')` with parameter bounds where appropriate (e.g., $\alpha \in (0, 1)$, $\sigma_e > 0$). Each function/gradient evaluation calls the worker callback (Section **??**). Warm-start from $(\tau^k, \tilde{\gamma}^k, \alpha^k, \sigma_e^k, \eta^k, \gamma_0^k)$.

Convergence tolerance: gradient norm $\leq 10^{-5}$. Maximum solver iterations: 100. In practice, L-BFGS on a 6-dimensional smooth problem converges in 20–50 function evaluations.

Closed-form updates. The variable μ is always updated by its closed-form expression (**??**), never by numerical optimization. Likewise ν is updated by the multiplier formula (**??**), not by optimization. The matrix inversion $(W + \rho_k I)^{-1}$ is $d_g \times d_g$ (e.g., 8×8) and negligible.

Penalty parameter. Initialize

$$\rho_0 = \frac{\operatorname{tr}(W)}{\dim(\mu)}.$$

This normalizes the penalty to the scale of the moment conditions. Update using the residual-balancing rule:

$$\rho_{k+1} = \begin{cases} 2\rho_k & \text{if } \|r^{k+1}\| > 10\, d^{k+1}, \\ \rho_k/2 & \text{if } d^{k+1} > 10\, \|r^{k+1}\|, \\ \rho_k & \text{otherwise,} \end{cases}$$

where $r^{k+1} = \mu^{k+1} - \bar{g}^{k+1}$ is the primal residual and $d^{k+1} = \rho_k \|\bar{g}^{k+1} - \bar{g}^k\|$ is the dual residual. Cap $\rho_k \leq 10^4$ to avoid ill-conditioning.

Convergence criteria. Stop when all of:

(i) primal residual: $\|r^{k+1}\| / \max(1, \|\mu^{k+1}\|) < 10^{-4}$;

(ii) dual residual: $d^{k+1} / \max(1, \|\nu^{k+1}\|) < 10^{-4}$;

(iii) parameter change: $\|\theta^{k+1} - \theta^k\| / \max(1, \|\theta^{k+1}\|) < 10^{-5}$.

Maximum outer iterations: 200.

Initialization.

(i) Run the pure micro MLE (ignoring the GMM penalty) to obtain initial $(\tau^0, \tilde{\gamma}^0, \{\delta_m^0, \tilde{q}_m^0\}_{m=1}^M)$, using the initialization in Section **??**. This can be distributed across markets via the profiled MLE in Section **??**.

(ii) Initialize $(\alpha^0, \sigma_e^0, \eta^0, \gamma_0^0)$ from the production-side moments conditional on the MLE initialization: regress $\ln R_j$ on $\ln(\hat{\tilde{Q}}_j L_j)$ to get α^0; use the A_j-free moment with $z_j = 1$ to get σ_e^0; regress δ_j^0 on $\ln w_j$ to get η^0; set γ_0^0 from the TFP normalization (**??**).

(iii) Set $\nu^0 = 0$.

(iv) Set $\mu^0 = (W + \rho_0 I)^{-1}(\rho_0 \bar{g}^0 - \nu^0) = (W + \rho_0 I)^{-1}(\rho_0 \bar{g}^0)$.

Weighting matrix. W is fixed throughout the augmented-Lagrangian iterations. If a two-step efficient GMM estimator is desired, run the full algorithm to convergence, construct $\hat{W}^{\text{opt}}$ from the residuals at $\hat{\theta}$, and re-run.

6.2.8 Communication Architecture

Memory model. Each worker holds only its own market's microdata D_m in memory. The coordinator holds only the current global parameters, the aggregate moment $\bar{g}$, and each market's moment contribution g_m (d_g floats per market). It never accesses raw microdata.

Communication per outer iteration. There are two phases of communication:

(a) *Market block (Steps 1–2):* One broadcast from coordinator, one return per worker. Each worker returns $(\delta_m^{k+1}, \tilde{q}_m^{k+1}, g_m^{k+1,k}, \ell_m^{k+1,k})$.

(b) *Global block (Step 4):* Multiple synchronous round-trips, one per function/gradient evaluation by the solver. Each round-trip: coordinator broadcasts 6 floats (trial global point), each worker returns (L_m, G_m, S_m, J_m) — totaling $1 + d_g + 2 + 5d_g = 6d_g + 3$ floats per worker. With $d_g = 8$: 51 floats per worker per round-trip.

With 20–50 solver iterations for the global block, the total per outer iteration is 20–50 synchronous round-trips. The binding cost is network latency, not bandwidth.

6.2.9 Potential Challenges and Tuning

Global solve latency. Each L-BFGS iteration in the global solve requires a synchronous round-trip to all workers. On a shared cluster with low-latency interconnect this is fast (the per-worker computation is a forward pass, not an optimization). Across data centers, latency could accumulate. Mitigation: (i) use a solver with fewer iterations (e.g., Newton-CG if Hessian-vector products are cheap); (ii) accept a loose tolerance for the global solve in early outer iterations and tighten as convergence approaches; (iii) batch multiple trial points per round-trip if the solver supports it.

Market subproblem conditioning. When ρ_k is large, the quadratic penalty term dominates the likelihood in (**??**), potentially making the subproblem ill-conditioned. The cap $\rho_k \leq 10^4$ mitigates this. If convergence stalls at large ρ, switch to a more robust solver (e.g., trust-region Newton) for the market subproblem. Monitor the number of inner iterations; if it grows rapidly with ρ, the penalty is too aggressive.

Slow outer convergence. If the primal residual $\|\mu - \bar{g}\|$ decreases slowly, possible causes include: (i) ρ_0 too small (the penalty is not pushing hard enough toward feasibility) — try $\rho_0 = 10$ or $\rho_0 = 100$; (ii) the global solve is too loose — tighten the gradient tolerance; (iii) the market subproblems are too inexact — tighten the inner tolerance. A diagnostic: if the primal residual oscillates, ρ is changing too rapidly; freeze ρ for 5–10 iterations.

Multiple inner sweeps. The algorithm as described performs one market sweep and one global solve per outer iteration. If outer convergence is slow, performing 2–3 market sweeps before the global solve (or vice versa) can help. This is closer to textbook augmented-Lagrangian theory, which allows multiple block-coordinate sweeps per multiplier update.

Sensitivity to initialization. The hybrid objective (**??**) may have local optima. The MLE-first initialization (Section **??**) is designed to start near the global optimum of the likelihood, so the GMM penalty is a refinement rather than a large correction. If the GMM moments pull parameters far from the MLE initialization, this suggests either a misspecified model or weak instruments. Diagnostic: compare the MLE-only and hybrid estimates; large discrepancies warrant investigation.

Straggler markets. The synchronous round-trip design means the slowest worker determines the iteration time. If markets have very different sizes (N_m varies by orders of magnitude), consider (i) assigning multiple small markets to one worker; (ii) using asynchronous updates for the market block (though this changes the convergence theory); (iii) profiling worker computation time and rebalancing.

Numerical stability of $\tilde{Q}_j^M(\sigma_e)$. Under Specification M, the truncated MGF $\exp(\sigma_e^2/2)\frac{\Phi(b-\sigma_e)-\Phi(a-\sigma_e)}{\Phi(b)-\Phi(a)}$ can overflow for large σ_e or extreme truncation bounds. Implement in log-space: compute $\log \tilde{Q}_j^M$ as a log-sum-exp over worker contributions. Use `jax.scipy.special.log_ndtr` for numerically stable $\log \Phi$.

6.3 GMM: Distributed Computation

The GMM objective cannot be profiled (quadratic form in a sum across markets): $Q(\theta) = (\sum_m S_m(\theta))' W (\sum_m S_m(\theta$ but gradients can be distributed:

$$\nabla_{\theta_m} Q = 2(\nabla_{\theta_m} S_m)'(W \textstyle\sum_m S_m), \qquad \nabla_{\theta_G} Q = 2(\textstyle\sum_m \nabla_{\theta_G} S_m)'(W \textstyle\sum_m S_m).$$

Open Question: Communication protocol: (i) two-round (workers send S_m, center broadcasts $W \sum S_m$, workers return Jacobian-vector products) vs. (ii) one-round (workers send S_m and full Jacobians)?

7 Standard Errors and Inference

Following Grieco et al. (Theorem 2), $\hat{\Gamma}_\theta^{-1}(\hat{\theta} - \theta^*) \xrightarrow{d} N(0, \mathcal{V}_\theta)$, where $\hat{\Gamma}_\theta = \hat{\Omega}_{\theta\theta} - \hat{\Omega}_{\theta\pi}\hat{\Omega}_{\pi\pi}^{-1}\hat{\Omega}_{\pi\theta}$. If $\hat{W}$ is chosen as in their Section 4.6, standard errors come from the inverse Hessian of (**??**); otherwise use the sandwich estimator.

Open Question: Incidental parameters with $2J$ fixed effects? Grieco et al. accommodate growing J via $M \to \infty$. In a single market, need $N \to \infty$ with J fixed.

8 Computational Notes

- Code in Python/JAX for autodiff and GPU acceleration.
- Memory: pre-computing $N \times J$ distance matrices requires ~8GB per market at $N_m \sim 10^6$, $J_m \sim 10^3$.
- Profiling out δ halves the outer parameters (from $2J$ to J) at the cost of running the contraction per evaluation.
- The standardized reparametrization separates the micro likelihood (depends on $\tilde{\gamma}, \tau, \{\tilde{q}_j, \delta_j\}$) from the macro penalty (additionally depends on $\gamma_0, \sigma_e, \alpha, \eta$). This enables a natural two-step procedure: estimate micro parameters first, then recover (γ_0, σ_e) from the production moments.

Open Question: Implicit function theorem for the gradient of the profiled objective vs. letting JAX differentiate through the fixed-point iteration?

Notation concordance with rcly_notes. The notes use different symbols: $\varphi \to \gamma_1$, $x_i \to v_i$, $a_i \to e_i$, $F_a \to G_e$, $c_j \to \bar{q}_j$, $V_j \to \delta_j$, $\gamma_{\text{notes}} \to \tau$, $\beta \to \alpha$, $S_j \to Q_j$. In the standardized reparametrization: $\tilde{q}_j = (\bar{q}_j - \gamma_0)/\sigma_e$ (Spec. A) or $(\ln \bar{q}_j - \gamma_0)/\sigma_e$ (Spec. M), $\tilde{\gamma} = \gamma_1/\sigma_e$.